for more :- http://www.UandiStar.org

Code No. N0522 **R07** **Set No.1**

IV B.Tech I Semester Regular Examinations, November, 2011

EMBEDDED SYSTEMS

(Common to Computer Science & Engineering and Information Technology)

Time: 3 hours **Max. Marks: 80**

Answer any FIVE Questions

All Questions carry equal marks

*****

1. a) Define the terms 'System' and an 'Embedded system'. Give the classification of embedded systems. [8]
   b) List the various complex systems available and explain their performance characteristics. [8]

2. a) Explain about TCON and TMOD function registers relevant to counters and timers of the 8051 microcontroller. [8]
   b) Tabulate the special function registers by making four columns as register, bit, primary function and bit addressable. [8]

3. a) List out and briefly explain the Boolean bit-level operations. [8]
   b) Write an assembly language program to OR the contents of port 1 and 2 and put the result in external RAM location 0100h.Write comment on this. [8]

4. a) Explain with suitable example, how to perform increment and decrement the contents of registers and RAM using relevant mnemonics. [8]
   b) Write an assembly language program to get hex data in the range of 00-FFh from the port 0 and convert it to decimal. Save the digits in R7, R6 and R5, where the LSB is in R7. Place comments on each line of code. [8]

5. With the aid of neat diagram the design approach of interfacing with keyboard with Display unit to 8051 microcontroller based embedded system. Write down the necessary program. [16]

6. What are the states of a task? Explain which is the entity controlling (Scheduling) the transitions from one state to another in a task? [16]

100% free SMS:- ON<space>UandiStar to 9870807070 for JNTU, Job Alerts, Tech News , GK News directly to ur Mobile

for more :- http://www.UandiStar.org

7. With suitable embedded system example, explain how laboratory tools are utilized for design using RTOS. [16]

8. What are the various Working modes of I^2C Bus and explain them with suitable Schematic diagrams. [16]

100% free SMS:- ON<space>UandiStar to 9870807070 for JNTU, Job Alerts, Tech News , GK News directly to ur Mobile

for more :- http://www.UandiStar.org

Code No. N0522 **R07** **Set No.2**

IV B.Tech I Semester Regular Examinations, November, 2011

EMBEDDED SYSTEMS

(Common to Computer Science & Engineering and Information Technology)

Time: 3 hours **Max. Marks: 80**

Answer any FIVE Questions

All Questions carry equal marks

*****

1. a) List the advanced microprocessors and microcontrollers used in embedded systems and present their role. [8]
   b) What are the functional circuits in a chip or core of a microcomputer? Explain them. [8]

2. a) Explain the following, with respect to the 8051 microcontroller register Organization. i) PC and DPTR ii) PSW [8]
   b) Discuss about external interrupts and software generated interrupts in 8051 microcontroller. [8]

3. Explain the following terms, relevant to programming the 8051 microcontroller.
   a) Lines of code
   b) Labels
   c) Instructions and
   d) Comments [16]

4. a) Explain about the jump and call program ranges with suitable diagram. [8]
   b) Write an assembly language program to add the byte in external RAM location 02CDh to internal RAM location 19h and put the result in to external RAM locations 00C0h(LSB) and 00C1h(MSB). Place comments on each line of code. [8]

5. Explain how the serial communication employed in 8051 microcontroller with suitable diagrams and necessary programming concepts. [16]

6. a) Explain the problem of sharing data by Multiple Tasks and Routines. [8]
   b) Explain how are Tasks different from functions and ISRs. [8]

7. a) Explain the Operating System units in an RTOS kernel. [8]
   b) Explain the tools availability in various RTOS. [8]

8. a) Explain the design approach of an Elevator Controller. [8]
   b) What is meant by Instruction level parallelism? Explain the same, in brief. [8]

100% free SMS:- ON<space>UandiStar to 9870807070 for JNTU, Job Alerts, Tech News , GK News directly to ur Mobile

for more :- http://www.UandiStar.org

Code No. N0522 **R07** **Set No.3**

IV B.Tech I Semester Regular Examinations, November, 2011

EMBEDDED SYSTEMS

(Common to Computer Science & Engineering and Information Technology)

Time: 3 hours **Max. Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks
*****

1. a) Explain about embedded processor for a complex system. [8]
   b) What are the techniques of power and energy management in an embedded system? Explain them in brief. [8]

2. a) Describe the various operating modes of the UART in brief. [8]
   b) Explain the following, relevant to serial data input/output in 8051 Microcontroller.
   i) Serial data interrupts ii) Data Transmission and Reception [8]

3. a) Explain how to perform testing programs on a single-board computer. [8]
   b) Write an assembly language program to add two 16-bit numbers. The numbers are 2E5Fh and A3B4h. Store the sum in R7 and R6. R6 should have the lower byte. Write comment on this. [8]

4. a) Explain the concept of interrupts and returns with suitable example. [8]
   b) Write an assembly language program to increment the contents of RAM locations 13h, 14h and 15h using indirect addressing. Place comments on each line of code. [8]

5. With the aid of neat diagram the design approach of interfacing Analog-to- Digital converter to 8051 microcontroller based embedded system.Write down the necessary program. [16]

100% free SMS:- ON<space>UandiStar to 9870807070 for JNTU, Job Alerts, Tech News , GK News directly to ur Mobile

for more :- http://www.UandiStar.org

6. a) With an example, explain the concept of deadlock situation during multitasking execution. [8]
   b) Explain the different scheduling methods used to control Task State Transition. [8]

7. a) Explain the hard real-time Scheduling Considerations. [8]
   b) Explain the terms "Semaphores and Queues" relevant to RTOS in an embedded system. [8]

8. a) Write notes on Internet-Enabled Systems. [8]
   b) Describe the applications and specifications of CAN Bus, in brief. [8]

100% free SMS:- ON<space>UandiStar to 9870807070 for JNTU, Job Alerts, Tech News , GK News directly to ur Mobile

for more :- http://www.UandiStar.org

Code No. N0522 **R07** **Set No.4**

IV B.Tech I Semester Regular Examinations, November, 2011

EMBEDDED SYSTEMS

(Common to Computer Science & Engineering and Information Technology)

Time: 3 hours **Max. Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks
*****

1. a) Explain the components of embedded system hardware. [8]
   b) Explain the importance of the following processors in embedded Systems.
   i) Digital signal processor ii) ASSP [8]

2. a) Explain the following, relevant to serial data input/output in 8051 Microcontroller.
   i) Serial data interrupts. ii) Data Transmission and Reception. [8]
   b) Draw the programming model of the 8051 microcontroller that has Collection of 8-and 16-bit registers and 8-bit memory locations. [8]

3. a) Explain the following, relevant to development tools using in assembly language programming.
   i) Simulators ii) Debuggers [8]
   b) What are the diamond decision elements? Discuss about them with simple 8051 decision instructions. [8]

4. a) Explain with suitable example, how to perform unsigned multiplication using relevant mnemonics. [8]
   b) Write an assembly language program to increment the contents of RAM locations 13h, 14h and 15h using indirect addressing. Place comments on each line of code. [8]

5. a) Explain the importance of TI flag and RI flag in serial data communication. [8]
   b) Write the 8051 assembly language program for detection and identification of key activation. [8]

100% free SMS:- ON<space>UandiStar to 9870807070 for JNTU, Job Alerts, Tech News , GK News directly to ur Mobile

for more :- http://www.UandiStar.org

6. a) Differentiate the terms 'OS' and 'Real Time OS' with respect to various Features. [8]
   b) Explain, which is the entity controlling (Scheduling) the transitions from one state to another in a task? [8]

7. a) What does embedded Software development mean? Explain in brief the different stages in the development and testing of an application. [8]
   b) Explain the terms "Semaphores and Queues" relevant to RTOS in an embedded system. [8]

8. a) What is meant by CAN Bus? List out the features of CAN Bus. [8]
   b) Give the features and a comparison of exemplary high performance ARM family of processors. [8]

100% free SMS:- ON<space>UandiStar to 9870807070 for JNTU, Job Alerts, Tech News , GK News directly to ur Mobile